

$\Rightarrow$ Space complexity of merge sort
 $=$ space required to store array + memory required for stack (recursion)
 $= O(n) + O(\log(n))$
 $= O(n + \log(n))$
 $= \underline{O(n)}$

$\Rightarrow$ Can we partition array in quick sort by moving only in one direction?



$\searrow$

$\checkmark \Rightarrow$ What sorting algo is used by sort in STL?

$\Rightarrow$ Hybrid

— { quick, heap,

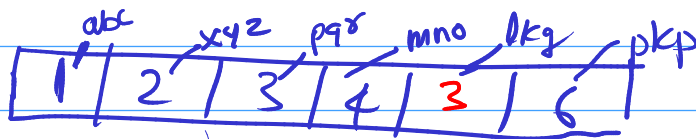
⇒ In-place sorting

→ Sorting does not require any extra auxiliary memory.

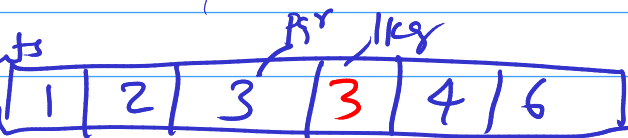
→ memory required is $O(1)$

	selection	✓
	Insertion	✓
	bubble	✓
	heap	✓
→	quick	X
	merge	X

⇒ stable sorting ✓ 3 | 5 | 7 | 3



Seq of elements with same val is as per original array. ✓



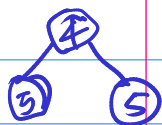
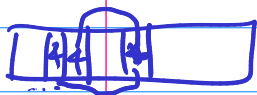
Stable sort

sorted?
Yes

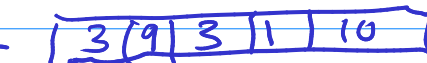
X



not a stable sort

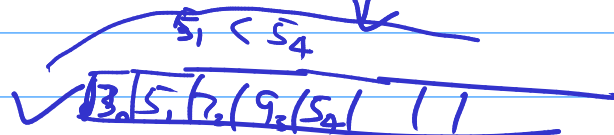


selection
bubble
insertion
heap
merge
quick

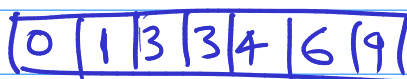
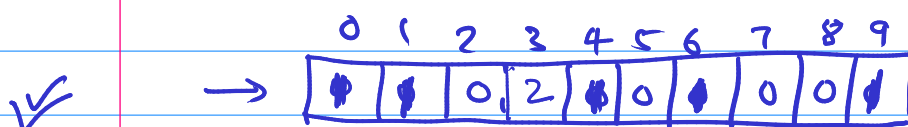


X
✓
✓
X
✓
X

could be made stable by clever idea



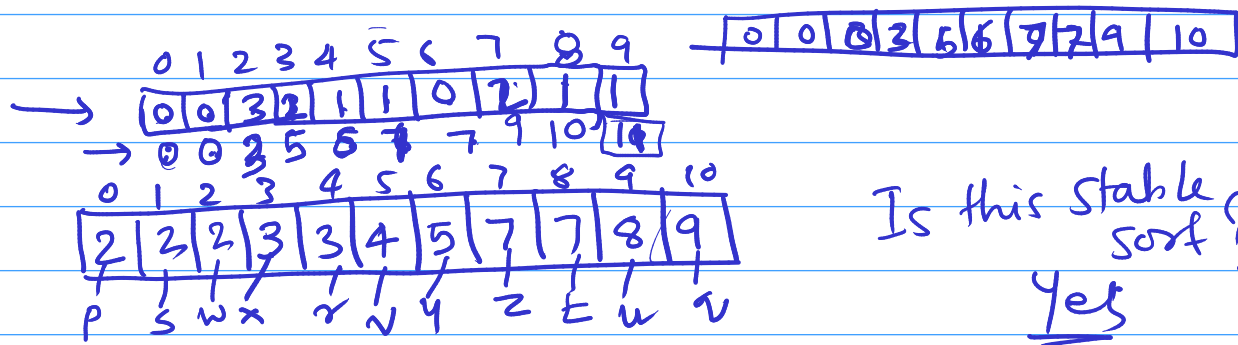
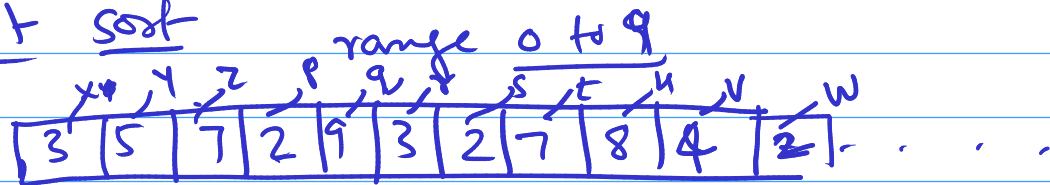
If you have 1 billion elements to sort
but range $0 \leq \text{element} \leq 9$
Input 3 | 6 | 9 | 4 | 0 | 1 | 3 | ...



Time: $O(n)$
Complexity: space: $O(\text{range})$

Drawback: Not a stable sort — can we solve this
: Does not work for large range

⇒ Count sort



Space complexity? $O(\text{range} + n)$

Time complexity? $O(n)$

⇒ Radix sort and bucket sort

⇒ Understand on your own

⇒ Try to observe similarity of radix sort with quick sort.